7	**(a)**		Magnetic flux density - force per unit current per unit length when conductor is placed perpendicular to the magnetic field	B2
			magnetic flux – product of area and magnetic flux density normal to the area	B1
	(b)		$\Phi = BA \sin\theta$	B1
	(c)	**(i)**	$\Phi = 1.8 \times 52 \times 10^{-2} \times 95 \times 10^{-2} \sin 90^{o}$	C1
			$= 0.89$ Wb	A1
		(ii)	As frame rotates, there is a rate of change of magnetic flux/cutting of flux,	B1
			By Faraday's law, (induced) e.m.f. is proportional to rate of change/cutting of (magnetic) flux (linkage)	B1
		(iii)	Sides PS and QR(both correct)	B1
		(iv)1	greatest rate of change of flux occurs when $\theta = 0°, 180^{o}$ and 360 or when plane of frame is parallel to field, so induced emf is maximum	B1
				B1
			least rate of change of flux occurs when $\theta = 90°$ and 270^{o} or when plane of frame is perpendicular to field, so induced emf is zero	B1
			Magnitude of e.m.f. determined by rate of change of flux	
		(iv)2	Since flux $\phi = BA \sin \theta = BA \sin\omega t$ which is sinusoidal	B1
			Induced emf = $d\phi/dt = BA\omega \cos \omega t$ is also sinusoidal	B1
		(iv)3	Max e.m.f. = $BA\omega = 0.89 \times 4\pi$	C1
			$= 11.2$ V	A1
		(iv)4	r.m.s current = $I_o/\sqrt{2} = (11.2/4.8)/\sqrt{2}z$	C1
			$= 1.646$ A	A1
			At slower angular speed, Rate of change of flux (linkage/cutting)	B1
			less peaks would be smaller amplitude Or maximum e.m.f. smaller.	B1

8	(a)	(i)	60 60 0	B1